



UNIVERSIDAD
PANAMERICANA

Materia: **Fundamentos de Programación en Paralelo**

Profesor: Dr. Juan Carlos López Pimentel

Evidencia:
Tareas y/o Ejercicios en Clase

Nombre de la actividad:
1p_threadsCreateFiles

Deadline: see in Blackboard.

Mode of work: in teams (check it in Blackboard)

Objective of the activity:

- Apply the knowledge of computer programming using threads to solve a problem requiring concurrency and synchronization.

Instructions:

You are going to simulate the value of a person in different moments of their life using different threads. To do that, you are going to use the formula of a person (P) found in the following video: <https://www.youtube.com/watch?v=q1kc-37C870>. It is not required to see the video to understand the problem, but it would be very valuable to do so.

Using Java language programming, develop the following:

Inputs:

- N: the number of threads to develop the work

- T: a maximum sleep timer in milliseconds.
- M: maximum number of iterations.

Process: N threads play the following. Each thread (worker) must create a file, within the file, write a phrase pinpointing the level of P that has in that moment. Each thread must iterate writing m times in the created file; at the end of each iteration each worker must sleep t milliseconds. P is the following formula:

$$P = (C+H) * A$$

- Variables C, H and A, must be generated randomly from 1 to 100.
- Variable m is a random number from 1 to M.
- Variable t is a random number from 10 to T.

Output:

- If we open any of the created files (while the thread is still iterating) we must see the different P annotations in the file.
- Generate an additional thread printing in the console the different states of each thread.
- When each thread finishes its work, it must display in the console the average P and you must display (according to your appreciation) if it is good, bad or else.

Requirements:

Req 1: Use of threads

Req 2: Use of sleep

Req 3: Use files

Req 4: Submit a pdf file with evidences accomplishing the instructions and requirements. In addition, it must include the source code.

Checklist:

Value	Characteristics to accomplish	Result		Observations
		Yes	Not	
	Penalty: 30% off for each of the following if it is not accomplished: Punctuality, delivering it in person, submit a pdf file			
About PDF				
30%	PDF contains evidence accomplishing Req 4.			
About the program				
5%	Inputs			
20%	Process			

20%	Output			
About comprehension				
25%	Answer to the questions of the teacher. This item is lost if the student is not present.			
100%	Score:			